

Procedural Modeling with Blender's Geometry Nodes

A workshop on taking advantage of the Geometry Nodes feature in Blender for procedural modeling

Jonathan Lampel
Orange Turbine, USA
jlampel@cgcookie.com

Jason van Gumster
Orange Turbine, USA
jason@cgcookie.com

ABSTRACT

Since about a year ago, the 3D program Blender has shipped with a set of procedural/parametric modeling tools called Geometry Nodes. While still fairly new and in development, the system has already unlocked several new ways for users to express their creativity and solve practical problems. It allows for the creation of new geometry from scratch, modification of existing geometry, and distribution of instances using a node graph. This presentation will cover those three major use cases, what's possible to create with the system, and demonstrate how to work with the tools by showing how to create a simple model generator.

KEYWORDS

Procedural modeling, Blender, Geometry Nodes, nodal workflow, workshop

ACM Reference Format:

Jonathan Lampel and Jason van Gumster. 2022. Procedural Modeling with Blender's Geometry Nodes: A workshop on taking advantage of the Geometry Nodes feature in Blender for procedural modeling. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH '22 Labs)*, August 07–11, 2022, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532725.3538516>

1 INTRODUCTION

Blender, the 3D digital content creation suite, has shipped with a set of procedural creation tools called Geometry Nodes since its 2.92 release over a year ago. This new feature gives users the ability to harness a powerful node-based interface for parametric and procedural modeling. While, as a feature, Geometry Nodes is fairly young and still under active development, the system has already unlocked new ways for users to express their creativity and solve practical problems. It allows for the creation of new geometry from scratch, modification of existing geometry, and distribution of instances using a node graph. This course covers those three major use cases. Participants will see what's possible to create with the system and learn how to work with the tools by building a simple model generator.

1.1 Requirements

This course is intended for beginner Blender users. No previous knowledge of Blender is required, but some understanding of the

basic concepts of 3D content creation is expected. At the end of the course, you will have a functional project file with an active example of a practical use case for Blender's Geometry Nodes. Since Blender itself is free and open source, you can continue working and exploring with Geometry Nodes after the course without needing to purchase any additional software.

Throughout the course, participants are encouraged to interrupt and ask questions relevant to the discussion at hand. One of the speakers will be giving the primary instruction while the other will be available to provide hands-on help if you are struggling with a particular step. The current release of Blender is version 3.2.

1.2 Course Description

The course begins with an introduction to Blender as well as a demonstration of what you should expect to learn. It starts with a description of what Blender is and a brief touch on its unique status as an open source 3D content creation suite. With that covered, you will open Blender for the first time and get comfortable with the basics of navigating its interface.

Rather than starting from scratch, there will be starter assets furnished to you, provided under a Creative Commons license that allows modification and reuse. Part of the introduction to Blender will involve pulling these assets into Blender and working with them. With the assets loaded, there will be a brief example of modeling in Blender with traditional, destructive mesh editing techniques. That approach will be contrasted with a more procedural, non-destructive approach using Blender's modifier stack.

The introduction to the modifier stack serves as the gateway to beginning to work with Geometry Nodes. After a demonstration of how to add a Geometry Nodes modifier to a mesh or curve object, a series of short examples will be shared to give participants a more concrete understanding of what can be done with Geometry Nodes in the hands of a talented artist.

The remainder of the course is devoted to helping you become that talented Geometry Node artist.

The process of discovering the capabilities available in Geometry Nodes starts with creating new geometry. By starting with a simple piece of plane geometry, you will learn how to transform that simple primitive into basic terrain. Then, using input nodes, you will see how to control the parameters and procedurally affect the look of the resulting terrain.

Participants will then learn how to distribute assets across their new surface. This process starts with understanding the difference between instanced geometry and realized geometry. With that understanding in place, you will see how to add instances of geometry to points on the terrain. Working with points naturally flows into a discussion on whether it's better to generate new points or convert existing geometry to points. Completing that discussion, you

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Labs, August 07–11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9367-6/22/08.

<https://doi.org/10.1145/3532725.3538516>

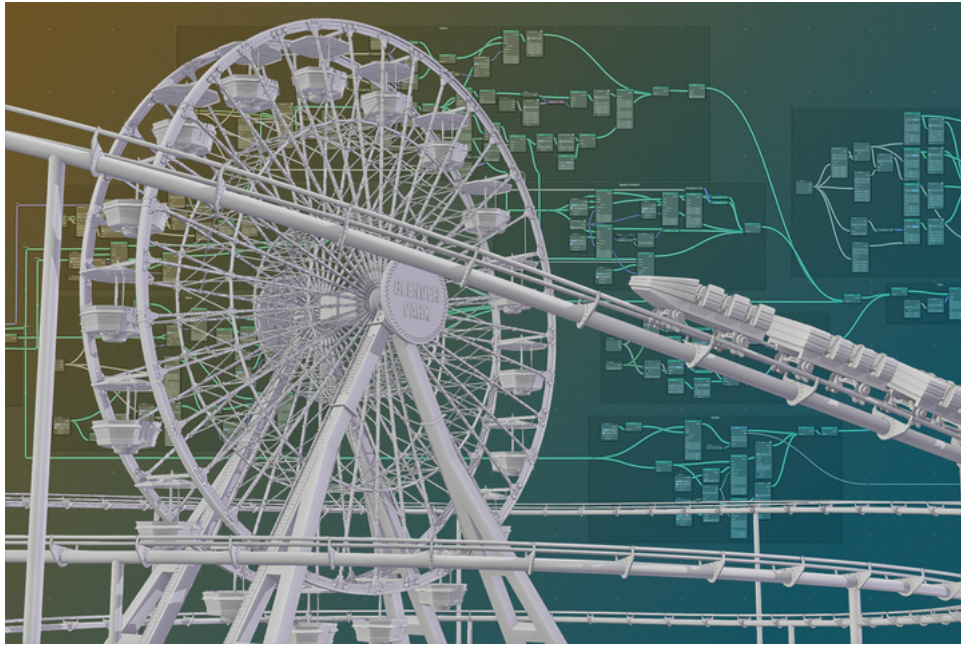


Figure 1: Example of a model created with Blender’s Geometry Nodes. Image by Jonathan Lampel. via CG Cookie.

will see how to use points to scatter some of the provided assets across their procedural terrains. This process also shows how to make the scattering and distribution feel more natural by imparting randomization to the size and rotation of instanced geometry.

Once the basics of distributing assets are understood, you will then be led into the somewhat more complex topic of creating dynamic effects. In essence, these are the tools to provide some simple rigging for their scenes using Geometry Nodes. This process starts with adding a new object to serve as an effector for the scene.

From there, you will be able to modify their node network such that the location of the effector will influence the location, rotation, scale, or any other attribute of instances in the scene.

With all the elements assembled in the scene, the course closes with participants learning how to animate the effector that they’ve added and then rendering their scene into a still image or video that they can share with others. At the very end, additional resources will be shared so you can continue refining your skills with Blender and Geometry Nodes.